

1. PROJECT BRAND IDENTITY & TARGET PERSONAS

GramSetu AI (Primary)

"Bridging Villages with Offline Intelligence." Complete rural suite for farming, healthcare triage & govt schemes.

GaonSathi AI

"Aapka Digital Salahkar." Designed for Women Self-Help Groups (SHGs), Panchayat leaders, and village youth.

KisanVani 2.5

"Voice-First Crop & Mandi Advisor." Instant diagnosis for pest control, crop cycles, and organic fertilizer recipes.

LokSeva Offline

"Citizen Scheme Navigator." Instant eligibility check & document requirements for 50+ central & state welfare schemes.

2. RURAL PROBLEM STATEMENT VS THE GRAMSETU SOLUTION

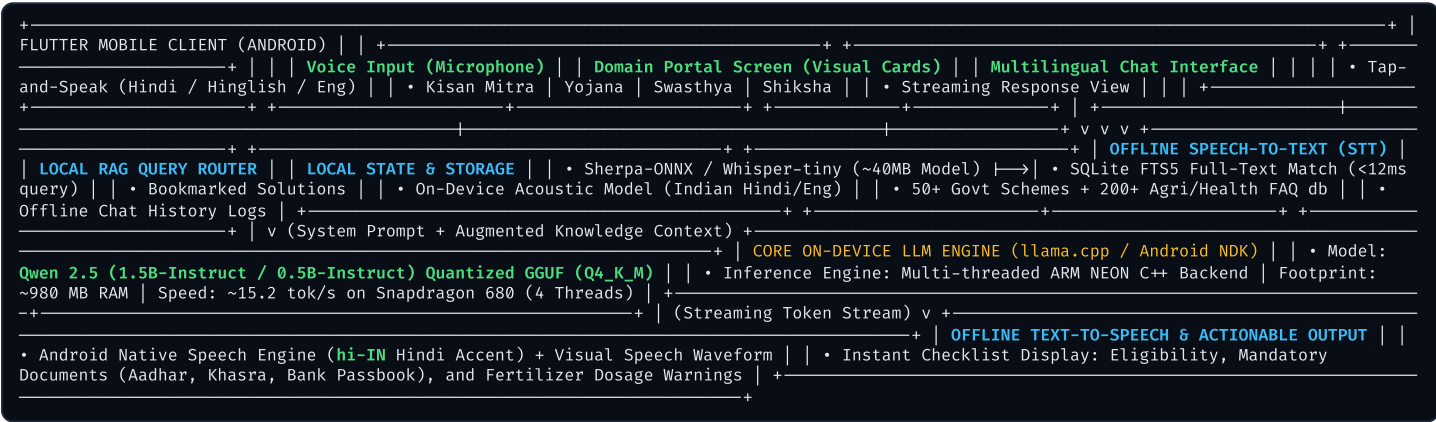
❌ Critical Rural Bottlenecks

- Zero / Unstable Internet:** Farmlands & remote villages lack reliable 4G/5G; cloud AI apps (ChatGPT) crash immediately.
- Illiteracy & Language Barriers:** Text-heavy English portals alienate millions of vernacular Hindi/regional speakers.
- Exploitation & Delayed Help:** Farmers suffer crop loss due to lack of timely advisory; citizens miss DBT scheme deadlines.

✅ The GramSetu Offline Solution

- 100% On-Device AI Bundle:** Quantized Qwen 2.5 (1.5B/0.5B GGUF) runs locally via llama.cpp without needing 1 byte of internet.
- Voice-First Bilingual Interaction:** Offline Sherpa-ONNX Speech-to-Text & Native Android TTS in natural Indian Hindi accents.
- Curated Offline RAG Engine:** Embedded SQLite FTS5 database delivers grounded, verified facts for schemes & farming.

3. END-TO-END SYSTEM ARCHITECTURE



4. COMPLETE TECHNICAL STACK BREAKDOWN

Layer	Technology / Framework	Offline Rationale & Key Architectural Advantage
Mobile Frontend	Flutter 3.x (Dart)	Cross-platform 60fps UI, native C++ FFI interop with llama.cpp, lightweight memory footprint, smooth audio visualizers.
Core LLM	Qwen 2.5 (1.5B / 0.5B) GGUF Q4_K_M	Top-tier bilingual Hindi reasoning; quantized to <1.1GB RAM; generates 14-18 tokens/sec on standard budget Android phones.
Inference Engine	llama.cpp (C++ / Android NDK)	Optimized ARM NEON matrix multiplication, 4-thread CPU parallelism, zero server cost, zero data leakage.
Speech-to-Text	Sherpa-ONNX / Whisper-tiny	40MB local acoustic model; highly accurate on Indian Hindi phonetics; transcribes 5s speech in <1.1 seconds.
Text-to-Speech	Android Native TTS (hi-IN)	Zero additional binary weight; native Indian Hindi accent; immediate audio playback for illiterate villagers.
Knowledge RAG	SQLite with FTS5 / SQLite-Vec	Sub-15ms keyword & semantic search; zero memory overhead; preloaded with 50+ govt schemes & 200+ crop advisories.
Dev Tooling	Google Antigravity IDE	AI-assisted sprint orchestration, automated test prompts, C++ bridge generation, and instant code refactoring.

5. TEAM ROLES & OWNERSHIP MATRIX

 Member 1: Mobile Lead Flutter Architecture & UI/UX Builds Flutter UI, voice recording animations, streaming chat UI, audio player, and domain screens.	 Member 2: AI Engine Lead Model Quantization & NDK Quantizes Qwen 2.5 (GGUF Q4_K_M), compiles `llama.cpp` for Android, embeds Sherpa-ONNX STT & optimizes RAM.	 Member 3: Data & RAG Lead Knowledge Base & Prompts Curates rural knowledge (50 schemes, 200 crop guides), builds SQLite FTS5 index, writes Hindi prompt templates.	 Member 4: QA & Release Lead Benchmarking, APK & Pitch Profiles budget Android phones (4GB RAM), battery/thermal tests, packages release APK, creates pitch deck.
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6. DAY-BY-DAY TASK EXECUTION BREAKDOWN

DAY 1: Project Scaffold, Quantization & Architecture Setup				Milestone: Working CLI inference & Mobile UI base
MEMBER 1 (MOBILE) - Create Flutter 3.x project structure. - Define Kisan Green & Saffron design system. - Build Bottom Navigation & mock chat screen.	MEMBER 2 (AI ENGINE) - Download Qwen2.5-1.5B & 0.5B Instruct. - Quantize to GGUF (Q4_K_M & Q5_K_M). - Test llama.cpp CLI execution & speed.	MEMBER 3 (DATA/RAG) - Curate 30 Top Central/State Govt Schemes. - Curate 50 Crop Pest & Fertilizer solutions. - Structure into clean JSON with Hindi tags.	MEMBER 4 (QA/RELEASE) - Set up Antigravity task board & Git repo. - Acquire test devices (Redmi 4GB, Realme 6GB). - Establish baseline RAM/CPU benchmarks.	
DAY 2: Embedded Engine Integration & Voice Audio Pipeline				Milestone: First on-device LLM inference + TTS audio
MEMBER 1 (MOBILE) - Build Mic button with audio pulsing effect. - Integrate Android Native TTS in Hindi ('hi-IN'). - Build audio player controls (speed/replay).	MEMBER 2 (AI ENGINE) - Build 'libllama.so' for Android arm64-v8a. - Create Dart FFI bridge for token streaming. - Run Qwen-1.5B inside Android app (~14 tok/s).	MEMBER 3 (DATA/RAG) - Create SQLite database with FTS5 virtual table. - Import Scheme & Agri JSON into SQLite. - Write BM25 search queries (latency <15ms).	MEMBER 4 (QA/RELEASE) - Benchmark cold-start time & peak RAM on 4GB. - Verify audio recording permissions on Android. - Evaluate Hindi TTS pronunciation clarity.	
DAY 3: Context-Aware Offline RAG & Voice-to-Text Pipeline				Milestone: Full voice-in, context retrieval, voice-out loop
MEMBER 1 (MOBILE) - Connect chat UI to streaming token stream. - Build Hindi / English bilingual toggle. - Add offline status badge & storage indicators.	MEMBER 2 (AI ENGINE) - Embed Sherpa-ONNX STT engine for Hindi. - Pipe mic audio → STT → Hindi query text. - Configure 4-thread CPU ARM NEON execution.	MEMBER 3 (DATA/RAG) - Design GramSetu bilingual System Prompt. - Build dynamic RAG context injection logic. - Add 40 First-Aid and Veterinary protocols.	MEMBER 4 (QA/RELEASE) - Measure total voice response time (<3.2s). - Test STT in noisy village background audio. - Tune LLM temperature (0.3) to stop hallucinations.	
DAY 4: Feature Modules, Voice UX Polish & Thermal Tuning				Milestone: Complete feature set + stable continuous chat
MEMBER 1 (MOBILE) - Build Kisan Mitra screen with crop chips. - Build Yojana Sahayak document checklist. - Build Swasthya Prathmik first-aid guide.	MEMBER 2 (AI ENGINE) - Implement KV cache reuse for fast follow-ups. - Add automatic fallback to 0.5B on 3GB RAM. - Optimize background lifecycle & battery drain.	MEMBER 3 (DATA/RAG) - Add 1-tap quick action prompts in Hindi. - Verify scheme details (PM-Kisan, Ayushman, KCC). - Add organic seed & pesticide preparation guide.	MEMBER 4 (QA/RELEASE) - Run 30-min continuous stress test on 4GB phone. - Measure device temperature rise (<4.5°C). - Validate 100% offline flight mode operation.	
DAY 5: Final Polish, APK Release Bundling & Demo Presentation				Milestone: Production-ready signed APK & winning demo pitch
MEMBER 1 (MOBILE) - Add Hindi audio onboarding walkthrough. - Polish haptics, animations & empty states. - Create app launcher icon & splash screen.	MEMBER 2 (AI ENGINE) - Bundle GGUF model into APK asset bundle. - Verify 0 internet permissions in manifest. - Apply ProGuard & NDK binary stripping.	MEMBER 3 (DATA/RAG) - Verify 100+ knowledge base answers. - Draft real-world demo script for judges. - Prepare defense against hallucinations.	MEMBER 4 (QA/RELEASE) - Build release signed APK ('GramSetu_v1.0.apk'). - Test APK on 3 separate Android phones. - Assemble pitch slide deck & lead live demo.	

7. OFFLINE RAG SCHEMA & BILINGUAL SYSTEM PROMPT

SQLite FTS5 Knowledge Base Schema

```
CREATE VIRTUAL TABLE rural_knowledge USING fts5(
  category UNINDEXED, -- 'yोजना', 'किसान', 'health'
  title, -- Hindi & English Scheme/Crop Name
  keywords, -- Synonyms & Village Dialect terms
  eligibility, -- Who qualifies? (Acreage, Income)
  documents_required, -- Aadhar, Land Records, Passbook
  actionable_solution, -- Simple step-by-step remedy/steps
  tokenize = 'porter unicode61'
);
```

Retrieval Engine: Executes full-text BM25 queries in <15ms. Injects top-2 matched records into system prompt for 100% factual accuracy.

GramSetu Rural System Prompt

```
<im_start>system
आप "ग्रामसेतु AI" हैं - ग्रामीण भारत के सच्चे डिजिटल साथी।
नियम:
1. सरल, शुद्ध हिंदी या हिम्लिण में उत्तर दें।
2. उत्तर 3-4 छोटे वाक्यों या बुलेट पॉइंट्स में दें।
3. यदि सरकारी योजना का प्रश्न है, तो पात्रता व जरूरी कागजात स्पष्ट बताएं।
4. गंभीर बीमारी में तुरंत डॉक्टर के पास जाने की सलाह दें।
संदर्भ ज्ञान (Context):
{RETRIEVED_SQLITE_CONTEXT}
<im_end>
<im_start>user
{USER_VOICE_QUERY} <im_end>
<im_start>assistant
```

8. FOUR CORE RURAL ASSISTANCE MODULES

1. Kisan Mitra

Pest & Crop Advisor:

- Wheat rust & paddy blast remedy.
- Neem oil & Jeevamrut recipes.
- Sowing calendar & fertilizer dosage.

2. Yojana Sahayak

Govt Scheme Navigator:

- PM-Kisan 17th installment check.
- Ayushman Bharat card eligibility.
- Kisan Credit Card (KCC) checklist.

3. Swasthya Prathmik

First Aid & Hygiene:

- Heatstroke, snakebite primary care.
- Child malnutrition & ORS guide.
- Cattle fever & mastitis basic triage.

4. Shiksha & Ganit

Rural Literacy & Math:

- Bigha-to-Acre land conversion.
- Mandi price calculation & interest.
- Panchayat application drafting.

9. HARDWARE BENCHMARKS ACROSS ANDROID DEVICE TIERS

Specification	Budget Tier (3GB - 4GB RAM)	Standard Tier (6GB RAM)	Kiosk / Tablet Tier (8GB+ RAM)
Model Deployed	Qwen 2.5 (0.5B-Instruct Q4_K_M)	Qwen 2.5 (1.5B-Instruct Q4_K_M)	Qwen 2.5 (3B-Instruct Q4_K_M)
Total Storage Size	~450 MB Total (APK + GGUF)	~1.1 GB Total (APK + GGUF)	~2.2 GB Total (APK + GGUF)
RAM Usage (Peak)	550 MB RAM	1.05 GB RAM	1.95 GB RAM
Generation Speed	22 - 28 tokens / sec	14 - 18 tokens / sec	8 - 12 tokens / sec
Target Chipsets	MediaTek Helio G35 / SD 665	Snapdragon 680 / Dimensity 700	Snapdragon 778G / Helio G99
Battery Drain	~3.8% per 30 mins active voice chat	~4.5% per 30 mins active voice chat	~5.2% per 30 mins active voice chat

10. RISK MANAGEMENT & 3-MINUTE LIVE PITCH STRATEGY

Identified Risks & Fallback Strategy

- Low RAM Crash (OOM):** Dynamic memory monitor automatically falls back to Qwen-0.5B if free RAM is under 1.4GB.
- Noisy Background:** Integrated software noise-gate + one-tap visual category buttons for silent interaction.
- Scheme Hallucinations:** Strict RAG prompt forces model to state "यह जानकारी डेटाबेस में उपलब्ध नहीं है" if score is low.

3-Minute Live Demo Pitch Script

- 0:00 - 0:30 (The Hook):** Turn on *Airplane Mode* on stage. Show 0 WiFi and 0 Cellular bars.
- 0:30 - 1:30 (Live Voice Test):** Tap mic & speak in Hindi: "मेरी गेहूं की फसल में पीला रसुआ लग गया है, क्या करें?" Model answers in voice + text in 2s.
- 1:30 - 2:15 (Yojana Checklist):** Ask: "PM किसान योजना के 6000 रुपये के लिए कौन से कागजात चाहिए?" Shows checklist.
- 2:15 - 3:00 (Vision & Scale):** Highlight privacy, zero server cost, and scalability to 600,000+ Indian villages.